Comparative Analysis of BERT, DistilBERT, and RoBERTa for Sentiment Classification on SST-2

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Abstract

This paper presents a comparative study of three transformer-based language models—BERT, DistilBERT, and RoBERTa—for binary sentiment classification on the Stanford Sentiment Treebank (SST-2) dataset. Each model was fine-tuned under identical experimental conditions using the Hugging Face Transformers library. Our experiments show that RoBERTa achieves the highest validation accuracy of 93.7%, followed by BERT at 93.2% and DistilBERT at 90.7%. However, DistilBERT trains in 73 minutes compared to 169 minutes for BERT and 806 minutes for RoBERTa, while using only 67M parameters (vs. 110M and 125M). These results provide practical guidance for selecting transformer models based on the trade-off between accuracy and computational efficiency.

I. Introduction

Sentiment analysis is a core task in natural language processing (NLP) that classifies text into positive or negative categories. The emergence of transformer-based pre-trained language models has significantly advanced the state of the art across NLP benchmarks, including the GLUE benchmark suite [1].

Among these models, BERT [2], DistilBERT [3], and RoBERTa [4] represent three distinct design philosophies: BERT established the bidirectional pre-training paradigm; DistilBERT compresses BERT through knowledge distillation for efficiency; and RoBERTa optimizes the pre-training procedure for maximum performance. While each model has been individually benchmarked, few studies provide controlled comparisons on the same task with identical hyperparameters and hardware.

This project conducts such a comparison on the SST-2 binary sentiment classification task. Our goal is to evaluate the trade-offs among classification accuracy, training time, and model size to provide practical recommendations for model selection in real-world applications. All experiments were conducted on an Apple Silicon Mac using the Hugging Face ecosystem, fulfilling the course requirement of integrating Hugging Face resources.

II. Method

A. Dataset

We use the SST-2 dataset from the GLUE benchmark [5], sourced from Hugging Face Datasets. SST-2 contains 67,349 training samples and 872 validation samples of movie review sentences, each labeled as positive (1) or negative (0). The test set labels are hidden, so all evaluations are performed on the validation set. The dataset is approximately balanced and sentences are short (mean ~19 words), making max_length=128 sufficient.

B. Models

We compare three pre-trained models loaded from the Hugging Face Model Hub:

BERT-base-uncased (110M parameters, 12 layers): The foundational bidirectional encoder model pre-trained with masked language modeling and next-sentence prediction [2].

DistilBERT-base-uncased (67M parameters, 6 layers): A compressed version of BERT obtained via knowledge distillation, designed to retain 97% of BERT's performance while being 40% smaller and 60% faster [3].

RoBERTa-base (125M parameters, 12 layers): An optimized variant of BERT trained with larger batches, more data, and dynamic masking, without the next-sentence prediction objective [4].

C. Experimental Setup

All models were fine-tuned using identical hyperparameters: learning rate 2e-5, batch size 16, weight decay 0.01, and 3 training epochs. We used the Hugging Face Trainer API with the AdamW optimizer and linear learning rate scheduling. Tokenization was performed with each model's corresponding tokenizer at max_length=128. Experiments ran on an Apple M-series chip with MPS acceleration. The evaluation metric is classification accuracy on the SST-2 validation set.

III. Results

A. Overall Performance

Table I summarizes the performance of all three models. RoBERTa achieved the highest best validation accuracy at 93.7% (Epoch 2), followed by BERT at 93.2% (Epoch 3) and DistilBERT at 90.7% (Epoch 3).

TABLE I. Model Performance Comparison on SST-2

Model	Params	Best Acc.	Train Time	Best Epoch
BERT-base	110M	93.2%	169 min	3
DistilBERT	67M	90.7%	73 min	3
RoBERTa-base	125M	93.7%	806 min	2

B. Per-Epoch Analysis

Table II shows the validation accuracy and loss at each training epoch. BERT and DistilBERT show monotonically increasing accuracy across epochs, while RoBERTa peaks at Epoch 2 and slightly decreases at Epoch 3, suggesting early signs of overfitting.

TABLE II. Per-Epoch Validation Results

Model	Ep.1 Acc	Ep.2 Acc	Ep.3 Acc	Ep.3 VLoss
BERT	92.2%	91.5%	93.2%	0.329
DistilBERT	90.1%	90.4%	90.7%	0.399
RoBERTa	92.8%	93.7%	93.5%	0.307

C. Efficiency Analysis

DistilBERT completed training in 73 minutes, approximately 2.3x faster than BERT (169 min) and 11x faster than RoBERTa (806 min). Despite having only 67M parameters compared to BERT's 110M and RoBERTa's 125M, DistilBERT's accuracy gap is modest: 2.5 percentage points below BERT and 3.0 points below RoBERTa. This makes DistilBERT attractive for resource-constrained deployment scenarios.

D. Error Analysis

Using the confusion matrix from RoBERTa (the best-performing model), we observe 392 true negatives, 427 true positives, 36 false positives, and 17 false negatives out of 872 validation samples. The model shows a slight tendency to misclassify negative sentences as positive (36 errors) compared to positive-as-negative errors (17). Examining misclassified examples reveals that ambiguous

or sarcastic sentences account for the majority of errors, which is consistent with known challenges in sentiment analysis.

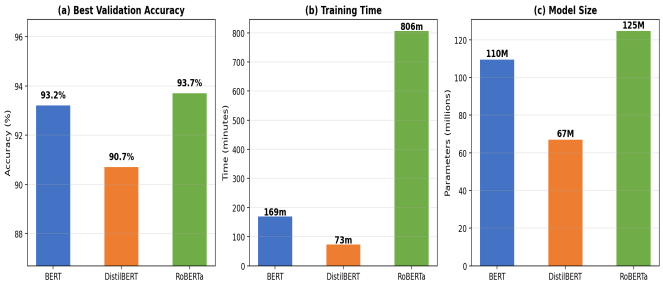

Fig. 1. Comparison of (a) validation accuracy, (b) training time, and (c) model size across the three models.

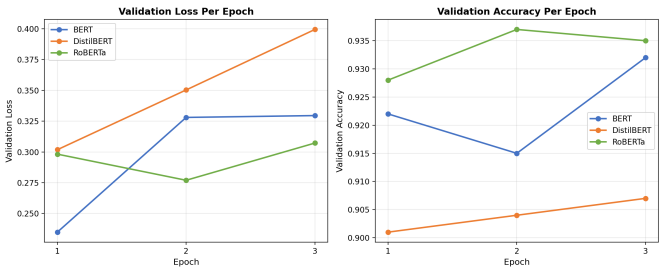

Fig. 2. Validation loss (left) and accuracy (right) across training epochs for all three models.

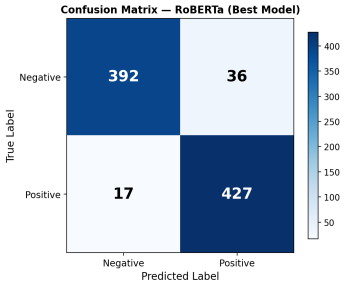

Fig. 3. Confusion matrix for RoBERTa on the SST-2 validation set.

IV. Conclusion

This study compared BERT, DistilBERT, and RoBERTa on the SST-2 sentiment classification task under identical fine-tuning conditions. RoBERTa achieved the highest accuracy (93.7%), validating the benefits of its optimized pre-training procedure. BERT provided a strong baseline (93.2%) with moderate training cost. DistilBERT, while 2–3 percentage points behind in accuracy, offered substantial efficiency gains with 2.3x faster training and 40% fewer parameters.

Our results are broadly consistent with published benchmarks, though slightly lower than the best reported figures (e.g., 94.0% for RoBERTa on SST-2 [6]), likely due to hardware constraints and hyperparameter choices. Limitations of this study include the lack of hyperparameter search, absence of additional metrics such as F1 score, and running on consumer hardware which limited batch sizes.

For practitioners, we recommend RoBERTa when accuracy is paramount and compute budget permits; BERT for a balance of performance and cost; and DistilBERT for latency-sensitive or resource-constrained deployments. Future work could explore learning rate tuning, longer training schedules, and attention visualization for deeper interpretability.

References

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